

Title: crop-yield-predictor
Project Report

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Fundamentals of AI Using Agriculture Data Set

Certificate

This is to certify that Krushna Kishor Patil from Indira College of Engineering and Management has successfully completed the project work titled "Crop-Yield-Predictor" as part of the 1-credit course “Fundamentals of AI using Agriculture Data Set”, offered under ANNAM.AI – Centre of Excellence, Ministry of Education, Government of India, at IIT Ropar. This project constitutes a component of the 30 hours of learning prescribed for the course and reflects the student’s independent effort in problem selection, method design, and implementation.

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1. Introduction and Motivation

Agriculture is the backbone of India's economy, employing nearly 54% of the workforce and contributing approximately 18% to the national GDP (2023–24 estimates). India is among the world's largest producers of rice, wheat, sugarcane, and pulses — yet paradoxically, millions of farmers face financial distress every year due to unpredictable yields, volatile market prices, and climate variability.

The ability to accurately predict crop yields before the harvest season is a transformative capability. With advance yield forecasts, farmers can:

- Make informed decisions about sowing patterns and crop selection
- Plan storage and logistics to minimise post-harvest losses
- Negotiate better prices with mandis and aggregators
- Access credit and insurance products calibrated to actual risk
- Help government agencies plan procurement, import/export, and buffer stocks

Traditional yield estimation relies on field surveys conducted by agricultural officers — a process that is expensive, slow, and prone to sampling bias. Machine learning offers a data-driven alternative that can synthesise historical records of weather, soil, area under cultivation, and irrigation to generate fast, scalable, and reasonably accurate forecasts.

This project focuses on building a supervised ML regression model that predicts district-level crop yield (kg/hectare) for major Kharif and Rabi crops across Indian states, using publicly available data from Government of India portals. The work is especially relevant to the Marathwada region (including Latur district, Maharashtra), where farmers growing soybean, tur (pigeon pea), and jowar face acute yield uncertainty due to erratic monsoons.

2. Problem Statement

2.1 Core Problem

Given historical agricultural data (area sown, production, rainfall, soil type, season, state/district) for the years 2001–2022, can we build a model that predicts the crop yield (kg/hectare) for a given crop, season, and region in the upcoming year?

2.2 Why This Problem Matters

India loses an estimated ₹90,000 crore annually to post-harvest losses and yield shocks. Yield prediction errors cascade into food price inflation, farmer debt, and in extreme cases, agrarian distress. A reliable forecasting tool, even with modest accuracy, provides significant economic value.

2.3 Formal Problem Definition

Component	Definition
Input Features (X)	State, District, Crop, Season, Area (ha), Annual Rainfall (mm), Fertiliser (tonnes), Pesticide (tonnes), Year
Target Variable (y)	Yield = Production (kg) / Area (ha)
Task Type	Supervised Regression
Evaluation Metric	RMSE, MAE, R ² Score
Success Criterion	R ² ≥ 0.85 on held-out test set
Output	Predicted yield in kg per hectare

2.4 Constraints and Scope

- Crops covered: Rice, Wheat, Maize, Soybean, Jowar, Cotton, Sugarcane, Tur (Arhar)
- States covered: All major agricultural states (33 states/UTs in dataset)
- Season focus: Kharif (June–November) and Rabi (November–April)
- Temporal scope: Training 2001–2019; Test 2020–2022

3. Dataset Understanding

3.1 Data Sources

The primary dataset is sourced from the data.gov.in portal — specifically the Crop Production Statistics dataset published by the Directorate of Economics and Statistics (DES), Ministry of Agriculture and Farmers Welfare, Government of India.

- Primary: data.gov.in — Crop Production Statistics (State/District, 2001–2022)
- Supplementary: IMD (India Meteorological Department) — district annual rainfall data
- Supplementary: FAI (Fertiliser Association of India) — state-wise fertiliser consumption
- Supplementary: India Pesticides dataset (CIB&RC portal)

3.2 Dataset Structure

Column	Data Type	Description	Example Value
State_Name	Categorical	Indian state or UT	Maharashtra

District_Name	Categorical	Revenue district	Latur
Crop_Year	Integer	Agricultural year	2018
Season	Categorical	Kharif / Rabi / Whole Year	Kharif
Crop	Categorical	Crop variety name	Soyabean
Area	Float	Area sown in hectares	120450.0
Production	Float	Output in metric tonnes	96360.0
Annual_Rainfall	Float	District rainfall in mm	778.5
Fertiliser	Float	Fertiliser used (tonnes)	45230.0
Pesticide	Float	Pesticide used (kg)	1230.0
Yield (derived)	Float	Production/Area (kg/ha)	800.0

3.3 Dataset Size and Split

Attribute	Value
Total Records (raw)	~246,000 rows
After cleaning & derived yield	~198,500 rows
Unique Crops	97
Unique States	33
Unique Districts	646
Year Range	2001 – 2022
Training Split (2001–2019)	~168,000 rows (85%)
Test Split (2020–2022)	~30,500 rows (15%)
Missing Value Rate (raw)	~6.4% (mostly Production column)

3.4 Key Characteristics and Observations

- Heavy right-skew in yield values — a small number of crops (sugarcane, potato) have very high yield/ha, while most cereals and pulses cluster at 500–2500 kg/ha. Log transformation applied.
- Seasonal imbalance: Kharif records account for 58% of data, Rabi 32%, and Whole Year / Zaid 10%.
- Maharashtra data: Latur and Osmanabad (Dharashiv) districts are strongly represented for soybean, tur, and jowar — the three most important crops in Marathwada.
- Rainfall data shows high variance — 2002 (severe drought) and 2019 (flood) years are outliers; model must handle these appropriately.
- Fertiliser and pesticide data is available at state level only and has been disaggregated to district level using area-proportional allocation (a known limitation).

4. Methodology

4.1 Overall Approach

The project follows a standard supervised ML pipeline: data ingestion → cleaning → feature engineering → model selection → training → evaluation → inference. Multiple regression algorithms are benchmarked before selecting the best performer.

4.2 Step-by-Step Methodology

Step	Activity	Rationale
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1	Data Collection & Merge	Combine crop statistics with rainfall, fertiliser, pesticide data on State+Year key
2	Data Cleaning	Remove rows where Production=0 or Area=0; impute missing rainfall with district median
3	Feature Engineering	Derive Yield = Production/Area; encode Season & Crop with Label Encoding; add Year ²
4	Outlier Removal	Remove records where Yield > 99th percentile (sugarcane skew); log-transform Yield
5	Train-Test Split	Temporal split: train on 2001–2019, test on 2020–2022 to simulate real forecasting
6	Scaling	StandardScaler on continuous features (Area, Rainfall, Fertiliser, Pesticide, Year)
7	Model Benchmarking	Compare Linear Regression, Decision Tree, Random Forest, Gradient Boosting (XGBoost)
8	Hyperparameter Tuning	GridSearchCV with 5-fold CV on training set for best model
9	Evaluation	RMSE, MAE, R ² on test set; feature importance analysis
10	Inference App	Simple Streamlit UI for farmers to get yield forecast

4.3 Feature Engineering Details

Target Variable

Yield (kg/ha) = (Production in tonnes × 1000) / Area in hectares. Log₁₀ transform applied to compress the heavy right tail; inverse-transform applied at prediction time.

Categorical Encoding

State and Crop are encoded using LabelEncoder (since tree-based models handle ordinal codes well). Season is one-hot encoded (Kharif=0, Rabi=1, Whole Year=2).

Temporal Feature

Year is kept as a numeric feature (2001=0 baseline shifted) to allow the model to learn long-term trend effects such as yield improvements from better seed varieties.

4.4 Model Selection Rationale

- Linear Regression: Baseline; good interpretability but poor with non-linear interactions.
- Decision Tree: Handles non-linearity but prone to overfitting; used as intermediate benchmark.
- Random Forest: Strong ensemble; handles outliers and missing data well; selected for final model.
- XGBoost: Gradient boosted trees; typically best performer but requires more tuning.

5. Implementation Details

5.1 Technology Stack

Tool / Library	Version	Purpose
Python	3.11	Primary programming language
Pandas	2.1.0	Data manipulation and cleaning
NumPy	1.26	Numerical computations

Scikit-learn	1.4.0	ML models, preprocessing, evaluation
XGBoost	2.0.3	Gradient boosted trees implementation
Matplotlib / Seaborn	3.8 / 0.13	Data visualisation
Streamlit	1.32	Interactive inference web app for farmers
Google Colab	—	Cloud compute (T4 GPU, 12GB RAM)
Joblib	1.3	Model serialisation and persistence

5.2 Implementation Pipeline

Step 1 — Data Loading and Merging

Three CSV files (crop_production.csv, district_rainfall.csv, fertiliser_state.csv) are loaded and merged using a hierarchical key of State + Year. District-level rainfall is joined on District + Year.

Step 2 — Cleaning

1. Remove rows with $\text{Area} \leq 0$ or $\text{Production} < 0$ (corrupt records): ~2,800 rows dropped
2. Impute missing rainfall values with the 5-year rolling district median
3. Drop rows where Yield > 99th percentile per crop (outliers from data entry errors)
4. Encode all categoricals using sklearn LabelEncoder; persist encoders for inference

Step 3 — Model Training

Final model: Random Forest Regressor with $n_estimators=300$, $max_depth=20$, $min_samples_leaf=4$, trained on $\log_{10}(\text{Yield})$. Training time on Colab: ~4 minutes.

Step 4 — Evaluation

Model evaluated on the held-out 2020–2022 temporal test set. Predictions are inverse-log-transformed before metric computation to report errors in native kg/ha units.

Step 5 — Inference App

A Streamlit app allows users to select State, District, Crop, Season, Area, Rainfall, and Year to receive a predicted yield range (point estimate $\pm$ 15% confidence interval based on training residual analysis).

5.3 Google Colab Link

The full implementation notebook is available at:

https://colab.research.google.com/drive/1iaan_H2OAkaJAHoZze-SB7dcudwj1r-U?usp=sharing

The repository containing all code, cleaned datasets, and the trained model pickle file is available at:

<https://github.com/Krishna009-pro/crop-yield-predictor>

6. Results and Discussions

6.1 Model Benchmark Comparison

Model	RMSE (kg/ha)	MAE (kg/ha)	R ² Score	Train Time
Linear Regression	642	498	0.61	< 1 sec
Decision Tree	583	412	0.68	< 1 sec
Random Forest (baseline)	389	281	0.83	3 min
Random Forest (tuned)	312	224	0.89	4 min
XGBoost (baseline)	371	265	0.85	5 min
XGBoost (tuned)	298	209	0.91	12 min

XGBoost (tuned) achieves the best R^2 of 0.91, meaning the model explains 91% of variance in crop yield. The Random Forest (tuned) is a close second with $R^2 = 0.89$. For the final deployment we selected the Random Forest as the primary model due to its faster inference and better interpretability for extension officers.

6.2 Feature Importance (Random Forest — Top 8 Features)

Rank	Feature	Importance Score	Interpretation
1	Crop (encoded)	0.284	Crop variety is the strongest predictor of yield
2	Annual Rainfall (mm)	0.198	Water availability directly drives productivity
3	State (encoded)	0.157	State-level agro-climatic zone and soil type captured
4	Area (ha)	0.112	Scale of farming; larger areas tend to have lower yield
5	Fertiliser (tonnes)	0.098	Input intensity; strong correlation with yield
6	Year	0.072	Captures technological progress over time
7	Season (encoded)	0.051	Kharif vs Rabi significantly differs in yield levels
8	Pesticide (kg)	0.028	Marginal but consistent positive correlation

6.3 Crop-wise Performance (Selected Crops, Test Set R^2)

Crop	R^2 Score	RMSE (kg/ha)	Notes
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Rice	0.92	187	Excellent — uniform cultivation practices
Wheat	0.93	142	Excellent — stable Rabi crop with irrigation
Soybean	0.87	198	Good — Marathwada crop; rainfall-sensitive
Cotton	0.81	241	Moderate — high volatility from pest pressure
Sugarcane	0.79	1,842	Moderate — very high absolute yield makes RMSE large
Tur (Arhar)	0.84	167	Good — pulse crop, rain-fed, performs well
Jowar	0.86	194	Good — important for dryland Marathwada farming

6.4 What We Learned

- Rainfall is a more powerful predictor than fertiliser in rain-fed districts like Latur and Osmanabad — confirming the real-world experience of farmers in Marathwada.
- Log-transforming the yield target significantly improved model performance ($R^2 + 0.07$) by reducing the influence of high-yield outliers.
- Temporal train-test splitting is crucial — random splitting overstates performance ($R^2 \sim 0.95$) because data from adjacent years leaks into training.
- Fertiliser usage data being available only at state level is the biggest data limitation; district-level data would likely reduce RMSE by 10–15%.

6.5 Challenges Faced

- Data inconsistency: Production column had ~6% missing values, concentrated in specific state-crop combinations. Imputation strategy significantly affects model quality for those groups.
- Seasonal disaggregation: Some records used 'Whole Year' as season, making it hard to align with Kharif/Rabi rainfall windows.
- Compute limitations: XGBoost hyperparameter search with 5-fold CV over large grids required ~45 minutes even on Colab GPU — managed by restricting the parameter grid.
- Interpretability vs accuracy tradeoff: Farmers and extension officers need explanations, not just numbers. SHAP values were computed to explain individual predictions but added deployment complexity.

7. Conclusion

This project successfully demonstrates that machine learning — specifically ensemble methods like Random Forest and XGBoost — can predict crop yields with high accuracy (R^2 up to 0.91) using publicly available Government of India datasets. The model captures the complex, non-linear interactions between rainfall, crop type, region, season, and input usage that determine agricultural productivity.

Key achievements of this project:

- Built an end-to-end ML pipeline from raw GOI data to a deployable prediction interface
- Achieved $R^2 = 0.89$ (Random Forest) and $R^2 = 0.91$ (XGBoost) on a realistic temporal test split
- Identified rainfall and crop variety as the dominant yield determinants, consistent with agronomic knowledge
- Developed a Streamlit application that a field-level extension officer can use without data science expertise

- Provided crop-wise performance breakdowns particularly relevant to Maharashtra's Marathwada region

The model has practical limitations — fertiliser data granularity, absence of soil health card data, and inability to capture micro-climate events like localised hailstorms. Future work should incorporate satellite remote sensing data (NDVI from Bhuvan/ISRO), soil organic carbon maps from the National Soil Health Card scheme, and district-level weather station data from IMD to further reduce prediction error.

With continued refinement, such a system could be integrated into the PM-KISAN portal or the Kisan Call Centre (KCC) platform, enabling proactive advisory services for India's 120 million smallholder farming families.

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A. Sample Dataset Records (First 10 Rows After Cleaning)

State	District	Year	Season	Crop	Area (ha)	Rainfall (mm)	Yield (kg/ha)
Maharashtra	Latur	2018	Kharif	Soyabean	120,450	778.5	803
Maharashtra	Osmanabad	2018	Kharif	Tur	62,340	778.5	612
Punjab	Ludhiana	2018	Rabi	Wheat	98,210	178.2	4,812
Uttar Pradesh	Varanasi	2019	Kharif	Rice	54,110	982.4	2,341
Rajasthan	Jaipur	2019	Rabi	Mustard	88,900	412.1	1,120
Andhra Pradesh	Guntur	2017	Kharif	Cotton	74,230	841.6	524
Karnataka	Belagavi	2020	Kharif	Maize	41,560	912.3	3,120
Bihar	Patna	2020	Rabi	Wheat	112,000	180.5	3,890
Gujarat	Surat	2021	Kharif	Sugarcane	28,770	1,234.0	78,410
Madhya Pradesh	Vidisha	2021	Kharif	Soyabean	95,400	890.2	1,015

B. Key Code Snippet — Model Training

```
# --- Feature Engineering ---
df['Yield'] = (df['Production'] * 1000) / df['Area']
df['log_Yield'] = np.log10(df['Yield'])
for col in ['State_Name', 'Crop', 'Season']:
```

```
le = LabelEncoder(); df[col] = le.fit_transform(df[col])

# --- Train-Test Split (Temporal) ---
train = df[df['Crop_Year'] <= 2019]
test = df[df['Crop_Year'] >= 2020]

# --- Model Training ---
rf = RandomForestRegressor(n_estimators=300,
                           max_depth=20, min_samples_leaf=4,
                           random_state=42, n_jobs=-1)
rf.fit(X_train, y_train_log)
y_pred = 10 ** rf.predict(X_test)
```

C. Evaluation Metrics Formulae

Metric	Formula	Interpretation
RMSE	$\sqrt{\sum (y_i - \hat{y}_i)^2 / n}$	Root Mean Square Error — penalises large errors; in kg/ha
MAE	$\sum y_i - \hat{y}_i / n$	Mean Absolute Error — average absolute deviation; robust to outliers
R ²	$1 - SS_{res} / SS_{tot}$	Coefficient of determination; 1.0 = perfect, 0 = predicts mean only

D. Future Scope

- Integrate NDVI (Normalised Difference Vegetation Index) from ISRO Bhuvan satellite data as a real-time crop health proxy
- Incorporate Soil Health Card (SHC) data from the SHC Portal to add soil pH, organic carbon, and micronutrient features
- Deploy as an API on NIC cloud for integration with PM-KISAN and Kisan Suvidha mobile apps

- Extend to district-level price prediction using AGMARKNET mandi prices to give farmers a revenue forecast, not just a yield forecast
- Build a time-series LSTM model to capture inter-annual dependency patterns that cross-sectional models miss

— End of Report —